AN INTEGRAL UNIVERSALLY INJECTIVE SURJECTIVE MORPHISM OF ALGEBRAIC SPACES INDUCES AN EQUIVALENCE OF SMALL ETALE TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1315-1326; proof span is lines 1328-1412, the complete author proof of 85 lines.

Editorial note. The proof contains no xymatrix display; the retained preamble is kept for the shared macro definitions.

Editorial note. The three cited cross-chapter results are prerequisites and are supplied as non-load-bearing navigation stubs.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Integral morphisms of algebraic spaces, universal homeomorphisms, the small etale site of an algebraic space, and descent of etale objects along a representable universal homeomorphism. The faithfulness, fullness and essential surjectivity arguments are retained.

Source reference. Original theorem etale-cohomology-theorem-topological-invariance: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1412–1412. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem spaces-descent-lemma-fpqc-universal-effective-epimorphisms: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1412–1412. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem spaces-morphisms-lemma-etale-universally-injective-open: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1412–1412. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem spaces-morphisms-section-universal-homeomorphisms: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1412–1412. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem spaces-theorem-presentation: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1412–1412. Referenced by the retained statement or proof but not part of the collected spans;

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is integral, universally injective and surjective. The functor*

$$V \longmapsto V_X = X \times_Y V$$

defines an equivalence of categories $Y_{spaces,étale} \to X_{spaces,étale}$.

Proof. The morphism f is representable and a universal homeomorphism, see Morphisms of Spaces, Section spaces-morphisms-section-universal-homeomorphisms.

We first prove that the functor is faithful. Suppose that V', V are objects of $Y_{spaces,étale}$ and that $a, b : V' \to V$ are distinct morphisms over Y. Since V', V are étale over Y the equalizer

$$E = V' \times_{(a,b),V \times_Y V,\Delta_{V/Y}} V$$

of a, b is étale over Y also. Hence $E \to V'$ is an étale monomorphism (i.e., an open immersion) which is an isomorphism if and only if it is surjective. Since $X \to Y$ is a universal homeomorphism we see that this is the case if and only if $E_X = V'_X$, i.e., if and only if $a_X = b_X$.

Next, we prove that the functor is fully faithful. Suppose that V', V are objects of $Y_{spaces,étale}$ and that $c : V'_X \to V_X$ is a morphism over X. We want to construct a morphism $a : V' \to V$ over Y such that $a_X = c$. Let $a' : V'' \to V'$ be a surjective étale morphism such that V'' is a separated algebraic space. If we can construct a morphism $a'' : V'' \to V$ such that $a''_X = c \circ a'_X$, then the two compositions

$$V'' \times_{V'} V'' \xrightarrow{\mathrm{pr}_i} V'' \xrightarrow{a''} V$$

will be equal by the faithfulness of the functor proved in the first paragraph. Hence a'' will factor through a unique morphism $a : V' \to V$ as V' is (as a sheaf) the quotient of V'' by the equivalence relation $V'' \times_{V'} V''$. Hence we may assume that V' is separated. In this case the graph

$$\Gamma_c \subset (V' \times_Y V)_X$$

is open and closed (details omitted). Since $X \to Y$ is a universal homeomorphism, there exists an open and closed subspace $\Gamma \subset V' \times_Y V$ such that $\Gamma_X = \Gamma_c$. The projection $\Gamma \to V'$ is an étale morphism whose base change to X is an isomorphism. Hence $\Gamma \to V'$ is étale, universally injective, and surjective, so an isomorphism by Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open. Thus Γ is the graph of a morphism $a : V' \to V$ as desired.

Finally, we prove that the functor is essentially surjective. Suppose that U is an object of $X_{spaces,étale}$. We have to find an object V of $Y_{spaces,étale}$ such that $V_X \cong U$. Let $U' \to U$ be a surjective étale morphism such that $U' \cong V'_X$ and $U' \times_U U' \cong V''_X$ for some objects V'', V' of $Y_{spaces,étale}$. Then by fully faithfulness of the functor we obtain morphisms $s, t : V'' \to V'$ with $t_X = \mathrm{pr}_0$ and $s_X = \mathrm{pr}_1$ as morphisms $U' \times_U U' \to U'$. Using that $(\mathrm{pr}_0, \mathrm{pr}_1) : U' \times_U U' \to U' \times_S U'$ is an étale equivalence relation, and that $U' \to V'$ and $U' \times_U U' \to V''$ are universally injective and surjective we deduce that $(t, s) : V'' \to V' \times_S V'$ is an étale equivalence relation. Then the quotient $V = V'/V''$ (see Spaces, Theorem spaces-theorem-presentation) is an algebraic space V over Y. There is a morphism $V' \to V$ such that $V'' = V' \times_V V'$. Thus we obtain a morphism $V \to Y$ (see Descent on Spaces, Lemma spaces-descent-lemma-fpqc-universal-effective-epimorphisms). On base change to X we see that we have a morphism $U' \to V_X$ and a compatible isomorphism

$U' \times_{V_X} U' = U' \times_U U'$, which implies that $V_X \cong U$ (by the lemma just cited once more).

Pick a scheme W and a surjective étale morphism $W \to Y$. Pick a scheme U' and a surjective étale morphism $U' \to U \times_X W_X$. Note that U' and $U' \times_U U'$ are schemes étale over X whose structure morphism to X factors through the scheme W_X. Hence by Étale Cohomology, Theorem etale-cohomology-theorem-topological-invariance there exist schemes V', V'' étale over W whose base change to W_X is isomorphic to respectively U' and $U' \times_U U'$. This finishes the proof. □